

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

CS2 - Risk Modelling and Survival Analysis
(Paper A)
May 2025

INDICATIVE SOLUTION

Solution 1: Correct Answer is Option C.

Let X be the temperatures in the city of Jaisalmer.

Let us define exceedance variable Y where $Y = X - 50$ as 50 is the threshold.
 $Y \mid X > 50$ is modelled using Generalised Pareto with parameters $\beta = 15, \gamma = 3$.
 When expressed as a two-parameter Pareto, $Y \mid X > 50 \sim \text{Pareto}(\gamma, \gamma\beta)$.

$$\begin{aligned} P(X > 52 \mid X > 50) &= P((X - 50) > 2 \mid X > 50) \\ &= P(Y > 2 \mid X > 50) \\ &= (\gamma\beta / (\gamma\beta + Y))^\gamma \\ &= (45 / 47)^3 \\ &= 0.877696 \end{aligned}$$

Climate risk actuary has selected $(150/3650) = 4.11\%$ of the data.

$$\begin{aligned} P(X > 52) &= P(X > 52 \mid X > 50) * P(X > 50) \\ &= 0.877696 * 0.0411 \\ &= 0.03607 \\ &= 3.61\%. \end{aligned}$$

[2]

Solution 2: Correct Answer is Option D.

A process is I (1) if it is not itself stationary but its first order difference is a stationary process.

A time series process is stationary when the roots of the characteristic polynomial are greater than 1.

A time series process would be I (1) when one of the roots of the characteristic polynomial is equal to 1 (the other root(s) being greater than 1). So that in the process of differencing, the root 1 gets eliminated and the first order differences are stationary.

Characteristic polynomial for X_t is:
 $1 - a\lambda - \frac{1}{2}\lambda^2 = 0$

$$\begin{aligned} \text{When } \lambda &= 1, \\ 1 - a - \frac{1}{2} &= 0 \\ a &= \frac{1}{2}. \end{aligned}$$

[2]

Solution 3: Correct Answer is Option A.

Probability that an unmarried person will always stay unmarried for the rest of his life indicates that the lifetime probability that the person remains unmarried till the very end (till he is alive) and finally goes into the dead state directly from the unmarried state.

$$\begin{aligned} P(\text{this happens in year 1}) &= 0.10 \\ P(\text{this happens in year 2}) &= 0.20 * 0.10 \\ P(\text{this happens in year 3}) &= 0.20^2 * 0.10 \\ P(\text{this happens in year } n) &= 0.20^{n-1} * 0.10 \end{aligned}$$

$$\begin{aligned} &\text{Required lifetime probability} \\ &= 0.10 + 0.20 * 0.10 + 0.20^2 * 0.10 + \dots + 0.20^{n-1} * 0.10 \end{aligned}$$

$$\begin{aligned} &\text{As } n \text{ tends to infinity,} \\ &\text{Required lifetime probability} \\ &= 0.10 / (1 - 0.20) \\ &= 0.125. \end{aligned}$$

[2]

Solution 4: Correct Answer is Option B.

$$n = 11$$

We first need to find the ratio r_1 .

$$\begin{aligned} &\text{Numerator} \\ &= 1 / (11-1) * ((\sum_{i=60}^{69} z_i * z_{i+1}) - (\sum_{i=60}^{70} z_i * \sum_{i=60}^{69} z_{i+1}) / 11) \\ &= 1 / 10 * (2.322794 - ((0.349668) * (0.349668 + 0.37165)) / 10) \\ &= 1/10 * 2.297572 \\ &= 0.2297572 \end{aligned}$$

$$\begin{aligned} &\text{Denominator} \\ &= 1 / 11 * (\sum_{i=60}^{70} z_i^2 - (\sum_{i=60}^{70} z_i)^2 / 11) \\ &= 1 / 11 * (4.62523 - 0.349668^2 / 11) \\ &= 1 / 11 * 4.1614119 \\ &= 0.419465 \end{aligned}$$

$$\begin{aligned} &r_1 \\ &= 0.2297572 / 0.419465 \\ &= 0.547739 \end{aligned}$$

$$\begin{aligned} &\text{t-test statistic} \\ &= \text{sqrt}(11) * r_1 \\ &= 1.816643 = 1.82. \end{aligned}$$

[3]

Solution 5: Correct Answer is Option E.

The two measures of distance would be equal only when:

$$\sqrt{(e - c)^2 + (f - d)^2} = |e - c| + |f - d|$$

$$\begin{aligned} &\text{Let } |e - c| = x \text{ and } |f - d| = y \\ &\sqrt{x^2 + y^2} = x + y \\ &\sqrt{x^2 + y^2} = \sqrt{x^2 + y^2 + 2xy} \end{aligned}$$

This is possible only when $2xy$ term is equal to 0. This is possible when either x or y or both are equal to 0.

In the context of this example, this is possible when either PE Ratio of ABC is equal to Cluster Centre A or ROI is equal or both PE Ratio and ROI are equal.

In all other cases, as $2xy$ term is non-zero, Euclidean distance will always be different from sum of absolute distance.

Statement 1 is not always true. Its true only when $2xy = 0$.

Statement 2 is never true. Euclidean distance can never be greater than sum of absolute distance.

Statement 3 and 4 – both are true.

[2]

Solution 6: Correct Answer is Option C.

Let us first derive the pseudo inverse function for t:

$$\Psi(t) = x = \ln\left(\frac{1 - \theta(1-t)}{t}\right)$$

$$\exp(x) = (1 - \theta(1 - t)) / t$$

$$t \exp(x) = 1 - \theta + \theta t$$

$$t(\exp(x) - \theta) = (1 - \theta)$$

$$t = (1 - \theta) / (\exp(x) - \theta)$$

$$\Psi^{-1}(t) = (1 - \theta) / (\exp(t) - \theta)$$

$$C[u, v]$$

$$= \Psi^{-1}(\Psi(u + v))$$

$$= \Psi^{-1}\left[\ln\left(\frac{1 - \theta(1-u)}{u}\right) + \ln\left(\frac{1 - \theta(1-v)}{v}\right)\right]$$

$$= \Psi^{-1}\left[\ln\left(\frac{1 - \theta(1-u)}{u} \cdot \frac{1 - \theta(1-v)}{v}\right)\right]$$

$$= \Psi^{-1}\left[\ln\left(\frac{(1 - \theta(1-u))(1 - \theta(1-v))}{uv}\right)\right]$$

$$= \frac{1 - \theta}{\frac{(1 - \theta(1-u))(1 - \theta(1-v))}{uv} - \theta}$$

$$= \frac{(1 - \theta)uv}{(1 - \theta(1-u))(1 - \theta(1-v)) - \theta uv}$$

$$= \frac{(1 - \theta)uv}{1 - \theta(1-v) - \theta(1-u) + \theta^2(1-u)(1-v) - \theta uv}$$

$$= \frac{(1 - \theta)uv}{1 - \theta(1-v + 1 - u + uv) + \theta^2(1-u)(1-v)}$$

$$= \frac{(1 - \theta)uv}{(1 - \theta) - \theta(1-u)(1-v) + \theta^2(1-u)(1-v)}$$

$$= \frac{(1 - \theta)uv}{(1 - \theta) - \theta(1-u)(1-v)(1 - \theta)}$$

$$= \frac{(1 - \theta)(1 - \theta(1-u)(1-v))}{uv}$$

$$= \frac{1}{(1 - \theta(1-u)(1-v))}$$

[3]

Solution 7: Correct Answer is Option D.

Let N denote the number of falls during the year.

$N \sim \text{Binomial}(1000, p)$

$$M_N(t) = (pe^t + 1 - p)^{1000}$$

Let X denote the cost of replacement of tri-pod legs. The distribution of X can be defined using the below table:

X	0	500	1000	1500
P(X) #	0.729	0.243	0.027	0.001

The probability of X also needs to be calculated using a Bin (3, 0.1) distribution as the probability of replacement for each leg is 0.1.

$$M_X(t) = 0.729 + 0.243e^{500t} + 0.027e^{1000t} + 0.001e^{1500t}$$

$$\begin{aligned} M_S(t) &= M_N(\ln M_X(t)) \\ &= (p M_X(t) + 1 - p)^{1000} \\ &= (0.729p + 0.243pe^{500t} + 0.027pe^{1000t} + 0.001pe^{1500t} + 1 - p)^{1000} \\ &= (1 - 0.271p + 0.243pe^{500t} + 0.027pe^{1000t} + 0.001pe^{1500t})^{1000} \end{aligned}$$

[3]

Solution 8: Correct Answer is Option E.

$$\begin{aligned} \text{Cov}(X_t, X_t) &= \text{Cov}(a + b e_t + c e_{t-1} + d e_{t-2}, a + b e_t + c e_{t-1} + d e_{t-2}) \\ &= b^2 \sigma^2 + c^2 \sigma^2 + d^2 \sigma^2 \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_t, X_{t-2}) &= \text{Cov}(a + b e_t + c e_{t-1} + d e_{t-2}, a + b e_{t-2} + c e_{t-3} + d e_{t-4}) \\ &= bd \sigma^2 \end{aligned}$$

Autocorrelation function at lag 2 is given by:

$$\begin{aligned} \rho_2 &= \text{Cov}(X_t, X_{t-2}) / \text{Cov}(X_t, X_t) \\ &= (bd) / (b^2 + c^2 + d^2). \end{aligned}$$

[2]

Solution 9: Correct Answer is Option A.

$$\begin{aligned} Y_t^2 - B_1 e_t^2 Y_{t-1}^2 &= 2(Y_t - B_1 e_t^2 Y_{t-1})u - (1 - B_1 e_t^2)u^2 + B_0 e_t^2 \\ Y_t^2 - 2Y_t u + u^2 &= e_t^2 (B_0 + B_1 Y_{t-1}^2 - 2B_1 Y_{t-1} u + B_1 u^2) \\ (Y_t - u)^2 &= e_t^2 (B_0 + B_1 (Y_{t-1} - u)^2) \\ Y_t &= u + /- e_t (B_0 + B_1 (Y_{t-1} - u)^2)^{0.5} \end{aligned}$$

$$E(Y_t) = E(u) + /- E(e_t (B_0 + B_1 (Y_{t-1} - u)^2)^{0.5})$$

Since e_t and Y_{t-1} are independent,

$$E(Y_t) = u + /- E(e_t) E((B_0 + B_1 (Y_{t-1} - u)^2)^{0.5})$$

Since $E(e_t) = 0$

$$\begin{aligned} E(Y_t) &= u + 0 \times E((B_0 + B_1 (Y_{t-1} - u)^2)^{0.5}) \\ E(Y_t) &= u. \end{aligned}$$

[2]

Solution 10: Correct Answer is Option B.

There are three states – 3% commission, 5% commission and 7% commission.

3% Commission:

1. P (Remain in 3% - Conservation Ratio of below 50%) = a_1
2. P (Move to 5% - Conservation ratio of anything above 50%) = $a_2 + a_3$
3. P (Move to 7% - not possible in one year) = 0

5% Commission:

1. P (Move to 3% - Conservation Ratio of below 50%) = a_1

2. $P(\text{Remain in 5\% - Conservation ratio from 50\% to 70\%}) = a_2$
3. $P(\text{Move to 7\% - Conservation ratio of above 70\%}) = a_3$

7% Commission:

1. $P(\text{Move to 3\% - not possible in one year}) = 0$
2. $P(\text{Move to 5\% - Conservation ratio below 70\%}) = a_1 + a_2$
3. $P(\text{Remain in 7\% - Conservation ratio of above 70\%}) = a_3$

So, the required one-year transition matrix is given below –

$$\begin{bmatrix} a_1 & a_2 + a_3 & 0 \\ a_1 & a_2 & a_3 \\ 0 & a_1 + a_2 & a_3 \end{bmatrix} \quad [2]$$

Solution 11: Correct Answer is Option A.

As per the constraints imposed on the Lee-Carter model,

1. the sum of all values of k_t shall be equal to 0.
2. the sum of all values of b_x shall be equal to 1.

By visual inspection it is clear, K1 and B1 satisfy these constraints and hence they are the correct representations of the respective parameters. [2]

Solution 12: Correct Answer is Option D.

Both processes exhibit the Markov property as the knowledge of past values apart from the current value is irrelevant in both the processes.

However, for process X_t , the increments I_t depend on the value of X_{t-1} . So, the process doesn't have independent increments.

But process Y_t has independent decrements as J_t does not depend on the values of Y_t .

Hence, statements 2, 3 and 4 are true. Statement 1 is false. [2]

Solution 13: Correct Answer is Option B.

The problem with cubic splines is that they can be too flexible, and may cause the model to include historical trend variations that are either short-term or past-specific, and which are not expected to recur in future. So, we introduce the smoothing parameter and the roughness penalty, to smoothen the cubic spline model. This is called penalised spline model.

If the smoothing parameter is made equal to 0, the roughness penalty will not be assigned and hence the mortality curve will be rough. It won't have a smoother projection. [2]

Solution 14: Correct Answer is Option E.

$$\text{Gini Index} = 1 - (p_1^2 + p_2^2)$$

[2]

Let us calculate Gini Index for Grades O to B.

Grade "X"	p_1	p_2	Gini Index
O	0.05	0.95	0.095
A+	0.15	0.85	0.255
A	0.30	0.70	0.420
B+	0.50	0.50	0.500
B	0.75	0.25	0.375

So, keeping the threshold at Grade O leads to minimum Gini index and hence the threshold shall be fixed at Grade O.

Solution 15: Correct Answer is Option C.

Let $P'_x(t)$ denote number of in-force policies aged x nearest birthday at time t .

Let $P_x(t)$ denote number of in-force policies aged x last birthday at time t .

We need the central exposed to risk E_x^C at age x nearest birthday at time t .

$$E_{46}^C = \frac{1}{2} * (P'_{46}(2022) + P'_{46}(2023)) + \frac{1}{2} * (P'_{46}(2023) + P'_{46}(2024))$$

No of policyholders with age 46 nearest birthday will be between ages 45.5 and 46.5. In terms of age last birthday, they will be between age last birthday 45 and age last birthday 46.

Assuming that birthdays are uniformly distributed throughout the year,

$$P'_{46}(2022) = \frac{1}{2} * (P_{46}(2022) + P_{45}(2022)) = 20,050$$

$$P'_{46}(2023) = \frac{1}{2} * (P_{46}(2023) + P_{45}(2023)) = 20,800$$

$$P'_{46}(2024) = \frac{1}{2} * (P_{46}(2024) + P_{45}(2024)) = 19,250$$

$$E_{46}^C = 0.5 * 20,050 + 20,800 + 0.50 * 19,250 = 40,450$$

$$\text{Force of mortality at age 46} = \mu_{46} = 2,760 / 40,450 = 0.0341 = 0.068232 = 6.82\%.$$

[2]

Solution 16:

Correct Answer is Option D.

$$C(u,v)$$

$$= u * v$$

$$= (1 - \exp(-\lambda x)) * (1 - \exp(-\mu y))$$

Now, using the given values of x and y at the 25th and 75th percentiles, we can substitute the values of $\hat{\lambda}_{MOP}$ and $\hat{\mu}_{MOP}$ for each of the options and check whether $C(u, v) = 0.25$ for the 25th percentile and $C(u, v) = 0.75$ for the 75th percentile.

$$\begin{aligned} \text{A. } C(u,v) &= (1 - \exp(-0.25*2.95)) * (1 - \exp(-0.75*1.31)) = 0.33 \\ C(u,v) &= (1 - \exp(-0.25*5.95)) * (1 - \exp(-0.75*7.00)) = 0.77 \end{aligned}$$

$$\begin{aligned} \text{B. } C(u,v) &= (1 - \exp(-0.50*2.95)) * (1 - \exp(-0.75*1.31)) = 0.48 \\ C(u,v) &= (1 - \exp(-0.50*5.95)) * (1 - \exp(-0.75*7.00)) = 0.94 \end{aligned}$$

$$\begin{aligned} \text{C. } C(u,v) &= (1 - \exp(-0.50*2.95)) * (1 - \exp(-0.25*1.31)) = 0.22 \\ C(u,v) &= (1 - \exp(-0.50*5.95)) * (1 - \exp(-0.25*7.00)) = 0.78 \end{aligned}$$

$$\begin{aligned} \text{D. } C(u,v) &= (1 - \exp(-0.25 \cdot 2.95)) \cdot (1 - \exp(-0.50 \cdot 1.31)) = 0.25 \\ C(u,v) &= (1 - \exp(-0.25 \cdot 5.95)) \cdot (1 - \exp(-0.50 \cdot 7.00)) = 0.75 \\ \text{E. } C(u,v) &= (1 - \exp(-0.50 \cdot 2.95)) \cdot (1 - \exp(-0.50 \cdot 1.31)) = 0.37 \\ C(u,v) &= (1 - \exp(-0.50 \cdot 5.95)) \cdot (1 - \exp(-0.50 \cdot 7.00)) = 0.92 \end{aligned}$$

[3]

So, the correct answer is Option D.

Solution 17: Correct Answer is Option A.

Let Z denote the conditional claim amount random variable representing the amounts paid by the reinsurer.

$$Z = X - M \mid X > M$$

Where

- X denotes the claims received by the direct life insurer, $X \sim \text{Pa}(\alpha, \lambda)$
- M is the retention limit

$$\begin{aligned} f_Z(Z) &= f_X(Z+M) / P(X > M) \\ &= (\alpha \lambda^\alpha / (\lambda + Z + M)^{\alpha+1}) / (\lambda^\alpha / (\lambda + M)^\alpha) \\ &= \alpha (\lambda + M)^\alpha / (\lambda + Z + M)^{\alpha+1} \end{aligned}$$

Thus, $Z \sim \text{Pa}(\alpha, \lambda + M)$

Equating the first order and second order moments of Z .

$$E(Z) = (\lambda + 39) / (\alpha - 1) = 6$$

$$\text{Var}(Z) = \alpha(\lambda + 39)^2 / ((\alpha - 1)^2 * (\alpha - 2)) = 49$$

$$\begin{aligned} 49 &= 6^2 * \alpha / (\alpha - 2) \\ 49\alpha - 98 &= 36\alpha \\ \alpha &= 98/13 = 7.538462 \end{aligned}$$

$$\begin{aligned} (\lambda + 39) / (98/13 - 1) &= 6 \\ \lambda &= (510 - 507)/13 = 3/13. \end{aligned}$$

We want to find the proportion of claims settled directly by the life insurer:

$$P(X \leq 39) = F_X(39) = 1 - (7.538462 / (7.538462 + 39))^{3/13} = 0.342992 = 34\%.$$

[4]

Solution 18: Correct Answer is Option E.

Statement 1 is true as if $\gamma = 1$ and $c = 1/\mu$, the mean failure time in years will be equal to μ under both the distributions.

Statement 2 is not true, as $\gamma > 0$ for a Weibull distribution. $\gamma = 0$ is invalid.

Statement 3 is also true.

$F(x) = 1 - \exp(-1/\mu * x^\gamma)$ under Weibull distribution.

$F(x) = 1 - \exp(-1/\mu * x)$ under Exponential distribution.

[2]

When $\gamma > 1$, the failure probabilities will be thus higher under Weibull distribution as compared to Exponential distribution.

Statement 4 is not true.

as $\gamma > 0$ for a Weibull distribution. $\gamma < 0$ is invalid.

So, statements 1 and 3 are true.

Solution 19: Correct Answer is Option E.

For checking whether the two processes are cointegrated, we need to check whether

- Both of them are I (1) process.
- A combination of these processes with the cointegrating vector is a stationary process.

$$\begin{aligned} X_t - Y_t &= aX_{t-1} + bY_{t-1} + e^x_t - (bX_{t-1} + aY_{t-1} + e^y_t) \\ &= a(X_{t-1} - Y_{t-1}) - b(X_{t-1} - Y_{t-1}) + (e^x_t - e^y_t) \\ &= (a-b)(X_{t-1} - Y_{t-1}) + (e^x_t - e^y_t) \end{aligned}$$

$$X_t - Y_t = (a-b)(X_{t-1} - Y_{t-1}) + (e^x_t - e^y_t)$$

$$\text{Let } Z_t = X_t - Y_t$$

$$Z_t = (a-b)Z_{t-1} + (e^x_t - e^y_t)$$

To check the stationarity of Z_t ,

Characteristic polynomial must be equated to 0.

$$\begin{aligned} 1 - (a-b)\lambda &= 0 \\ \lambda &= 1 / (a-b) \end{aligned}$$

Only when $|a - b| < 1$, then the absolute value of the root of the characteristic polynomial is greater than 1.

$|a - b| < 1$ translates to $(a - b) < 1$ or $(a - b) < -1$ i.e. $(b - a) > -1$.

So, the right answer is both Options (A) and (C).

[2]

Solution 20: Correct Answer is Option E.

For the given Markov Chain, the transition matrix showing one time unit's transition probabilities is as follows:

$$\begin{bmatrix} & \text{Space} & \text{Math} & \text{Stat} \\ \text{Space} & 0.5 & 0.25 & 0.25 \\ \text{Math} & 0 & 1 & 0 \\ \text{Stat} & 0 & 0 & 1 \end{bmatrix}$$

For arriving at the stationary distribution, we have –

$$\pi_{\text{SPACE}} = 0.5 * \pi_{\text{SPACE}} \dots\dots\dots (1)$$

$$\pi_{\text{MATH}} = 0.25 * \pi_{\text{SPACE}} + \pi_{\text{MATH}} \dots\dots\dots (2)$$

[2]

$$\pi_{\text{STAT}} = 0.25 * \pi_{\text{SPACE}} + \pi_{\text{STAT}} \dots\dots\dots(3)$$

$$\text{From equations (1), (2) and (3) it is clear that } \pi_{\text{SPACE}} = 0 \dots\dots\dots (4)$$

$$\text{Further, } \pi_{\text{SPACE}} + \pi_{\text{MATH}} + \pi_{\text{STAT}} = 1 \dots\dots\dots (5)$$

So, we deduce from (4) and (5) that

$$\pi_{\text{MATH}} + \pi_{\text{STAT}} = 1$$

So, this Markov Chain does not have a unique stationary distribution.

Any values of π_{MATH} and π_{STAT} which add up to 1, and $\pi_{\text{SPACE}} = 0$ can be called a stationary distribution.

Hence all options (A) to (D) are true. Right answer is Option (E).

Solution 21: Correct Answer is Option C.

The generator matrix in respect of the model is given below for IF “In-Force”, LP “Lapsed” and DH “Death” states:

$$\begin{bmatrix} & \text{IF} & \text{LP} & \text{DH} \\ \text{IF} & -0.12 & 0.10 & 0.02 \\ \text{LP} & 0 & 0 & 0 \\ \text{DH} & 0 & 0 & 0 \end{bmatrix}$$

$$d/dt p_{\text{IF IF}} = p_{\text{IF IF}} * \mu_{\text{IF IF}} = -0.12 * p_{\text{IF IF}}$$

$$d/dt p_{\text{IF IF}} / p_{\text{IF IF}} = -0.12$$

$$d/dt \log_e(p_{\text{IF IF}}) = -0.12$$

Integrating on both sides with respect to t,

$$\log_e(p_{\text{IF IF}}) = -0.12t + c$$

$$\text{At time 0, } p_{\text{IF IF}} = 1$$

$$\log_e(1) = 0 + c$$

$$c = 0$$

$$\log_e(p_{\text{IF IF}}) = -0.12t$$

$$p_{\text{IF IF}} = \exp(-0.12t). \dots\dots\dots (1)$$

We are interested in finding the value of $p_{\text{IF DH}}$, as only then we can find out the expected cost of claims in the first year.

$$\begin{aligned} d/dt p_{\text{IF DH}} &= p_{\text{IF IF}} * \mu_{\text{IF DH}} + p_{\text{IF DH}} * \mu_{\text{DH DH}} = p_{\text{IF IF}} * 0.02 + 0 \\ &= p_{\text{IF IF}} * 0.02 \\ &= \exp(-0.12t) * 0.02 \dots\dots\dots \text{From (1)} \end{aligned}$$

$$d/dt p_{IF\ DH} = \exp(-0.12t) * 0.02$$

Integrating both sides with respect to t,

$$p_{IF\ DH} = -0.02 / 0.12 * \exp(-0.12t) + c$$

At time 0, $p_{IF\ DH} = 0$. So, $c = 0.02/0.12$

$$p_{IF\ DH} = -0.02 / 0.12 * \exp(-0.12t) + 0.02/0.12$$

At time $t=1$,

$$\begin{aligned} p_{IF\ DH} &= -0.02 / 0.12 * \exp(-0.12) + 0.02/0.12 \\ &= 0.018847 \end{aligned}$$

So, expected cost of claims in the first year

$$= 1 \text{ crore} * 0.0018847$$

$$= \text{INR } 1,88,466.$$

[4]

Solution 22: Any Option is correct.

Durations and outcomes are shown in the table below:

Patient	Died / Censored	Duration
Donald	Died	5
Cherry	Died	12
Andy	Censored	14
Henry	Died	24
Fred	Censored	30
George	Censored	34
Emily	Censored	39
Bernard	Censored	56

Let's calculate the Kaplan-Meier estimate of the survival function:

t_j	n_j	d_j	c_j	$\lambda_j = d_j/n_j$	$\lambda_j / (n_j - d_j)$
0	8	0	0	0	0
5	8	1	0	1/8	1/56
12	7	1	1	1/7	1/42
24	5	1	0	1/5	1/20

The Kaplan-Meier estimate of the survival function is given by:

t	S(t)	Var (S(t)) *
$0 \leq t < 5$	1	
$5 \leq t < 12$	0.875	
$12 \leq t < 24$	0.750	$0.750^2 * (1/56 + 1/42)$ $= 0.0234$

* $Var(S(t))$ is calculated using the Greenwood's Formula

Survival probability of surviving for 3 weeks
 $= S(21) = 0.750$

90% Confidence Interval for $S(21)$
 $= 0.750 \pm 1.645 * \sqrt{0.0234}$
 $= (0.4982, 1.0000]$.

[4]

Solution 23: Correct Answer is Option A.

Statement 1 is Type I censoring as the study stops at 2024.

Statement 2 is Type II censoring because the study for a particular portfolio stops after defaults breaching the limit of 20% of the total value of loans.

Statement 3 is Informative censoring because the individuals excluded from analysis are the ones with high CIBIL scores.

Statement 3 is Non-Informative censoring as the analysis excludes the pandemic period for all borrowers, all portfolios.

[2]

Solution 24: Correct Answer is Option E.

The hazard function for rate of unsubscribing is given by –

$$\lambda(t, Z) = \lambda_0(t) * \exp(\beta_1 Z_1 + \beta_2 Z_2 + \beta_3 Z_3)$$

For 2023,

$$1 - 65\% = 35\% = \exp[-\exp(\beta_1 + \beta_2) \int \lambda_0(t) dt]$$

For 2024, the first two parameters remain the same.

In respect of the third parameter i.e. cost,

GETFLIX fees = 130% of 2023 GETFLIX fees.

Fees of nearest competitor
 $= 80\% * 105\%$ of 2023 GETFLIX fees
 $= 84\%$ of 2023 GETFLIX fees

So, GETFLIX fees in terms of the market leader
 $= 130\% / 84\%$
 $> 120\%$ of the market leader

Subscription retention rate for the platform in 2024
 $= 1 - \exp[-\exp(\beta_1 + \beta_2 + \beta_3) \int \lambda_0(t) dt]$
 $= 1 - \exp[-\exp(\beta_1 + \beta_2) * \exp(\beta_3) * \int \lambda_0(t) dt]$
 $= 1 - \exp[-\exp(\beta_1 + \beta_2) * \int \lambda_0(t) dt * \exp(\beta_3)]$
 $= 1 - \{\exp[-\exp(\beta_1 + \beta_2) * \int \lambda_0(t) dt]\}^{\exp(\beta_3)}$
 ≈ 0.57

So, subscription retention rate for 2024 is 57%.

[3]

Solution 25: Correct Answer is Option **D**.

Graph 1 is monotonically decreasing hazard.

Graph 2 is log-logistic hazard.

Graph 3 is exponential hazard.

Graph 4 is constant hazard.

[2]

Solution 26: Correct Answer is Option **E**.

Under existing scenario,

Total claim amounts

$$= 1/3 * 0 + 1/3 * (1/2 * 1 + 1/2 * 2) + 1/3 * (1/4 * 2 + 1/2 * 3 + 1/4 * 4) \\ = 1.50$$

$$\text{Premium} = 120\% \text{ of } 1.50 = \text{INR } 1.80.$$

Let us consider profits under the present situation

$$= \text{Loadings} = 1.80 - 1.50 = \text{INR } 0.30$$

With proposed policy excess,

Total claim amounts

$$= 1/3 * 0 + 1/3 * (1/2 * (1-E) + 1/2 * (2-E)) + 1/3 * (1/4 * (2-2E) + 1/2 * (3-2E) + 1/4 * (4-2E)) \\ = 1/3 * (1.5 - E) + 1/3 * (3 - 2E) \\ = 1.5 - E$$

$$\text{Premium} = 1.80 - 0.10 = 1.70$$

Profits

$$= 1.70 - 1.5 + E \\ = 0.20 + E$$

We want the profits to be higher in case of policy excess.

$$0.20 + E > 0.30$$

$$E > 0.10$$

$$E > 0.10.$$

[2]

Solution 27: Correct Answer is Option **A**.

N is the random variable representing number of claims. Let NR be the random variable representing number of claims settled by the reinsurer.

Let us define NR in terms of N.

$$NR = 0 \text{ if } N = 0 \text{ or } 1 \\ NR = (N - 1) \text{ if } N > 1$$

$$E(NR)$$

$$\begin{aligned}
&= 0 + 1 * P(NR=1) + 2 * P(NR = 2) + 3 * P(NR=3) + \dots \\
&= 0 + 1 * P(N=2) + 2 * P(N=3) + 3 * P(N=4) + \dots \\
&= 1 * p * (1-p)^2 + 2 * p * (1-p)^3 + 3 * p * (1-p)^4 + \dots \\
&= p * [(1-p)^2 + 2 * (1-p)^3 + p * (1-p)^4 + \dots] \\
&= p * [(1-p)^2 + (1-p)^3 + (1-p)^4 + \dots \\
&\quad + (1-p)^3 + (1-p)^4 + \dots \\
&\quad + (1-p)^4 + \dots \\
&\quad + \dots] \\
&= p * [(1-p)^2 / (1-(1-p)) + (1-p)^3 / (1-(1-p)) + (1-p)^4 / (1-(1-p)) + \dots] \\
&= p * [(1-p)^2 / p + (1-p)^3 / p + (1-p)^4 / p + \dots] \\
&= p/p * [(1-p)^2 + (1-p)^3 + (1-p)^4 + \dots] \\
&= 1 * (1-p)^2 / (1-(1-p)) \\
&= (1-p)^2 / p
\end{aligned}$$

$$E(NR) = (1-0.50)^2 / 0.50 = 0.50$$

S_R denote the aggregate claim amounts paid by the reinsurer.

$$\begin{aligned}
S_R &= 0.50X_1 + 0.50X_2 + \dots + 0.50X_{NR} \\
&= 0.50 (X_1 + X_2 + \dots X_{NR})
\end{aligned}$$

$$\begin{aligned}
E(S_R) &= 0.50 E(X_1 + X_2 + \dots X_{NR}) \\
&= 0.50 * E(NR) * E(X_i) \\
&= 0.50 * 0.50 * 5000 \\
&= \text{INR } 1,250.
\end{aligned}$$

[4]

Solution 28: Correct Answer is Option D.

Let us consider each statement.

1. Hazard rate for exponential distribution $= f(x) / (1 - F(X)) = \lambda \exp(-\lambda x) / \exp(-\lambda x) = \lambda$. Hence the hazard rate is a constant λ and not an increasing function of X . Hence the statement is not correct.
2. Mean residual life for an exponential distribution $= 1/\lambda * \exp(-\lambda x) / \exp(-\lambda x) = 1/\lambda$. Hence the mean residual life is a constant $1/\lambda$ and not a decreasing function of X . Hence the statement is not correct.
3. Existence of moments $E(X^k)$ for all positive integers k indicates that the exponential distribution has a light tail. It does not indicate that the distribution has a heavy tail. Hence the statement is not correct.
4. The mean residual life is the multiplicative inverse of hazard rate in case of an exponential distribution. This is a correct statement as λ and $1/\lambda$ are multiplicative inverses of each other.
5. If we compute the limiting density ratio of $\lambda=2$ against $\lambda=3$, then as x tends to infinity, the limiting density ratio tends to infinity (and not zero) and this indicates that $\lambda=2$ has a thicker tail. Hence this statement is not correct.

So, the only correct statement is Statement 4.

[3]

Solution 29: Correct Answer is Option B.

Let the transition rate be defined as $f(t)$ as it is a function of time t .

Probability that the process remains in state 0 from time 0 to s
 $= p_{00}(0, s) = \exp\left(-\int_0^s f(t)dt\right)$

So, $\exp(-s^2) = \exp\left(-\int_0^s f(t)dt\right)$

$$s^2 = \int_0^s f(t)dt$$

$f(t)$ is that function which when integrated from 0 to s gives us s^2 .

Looking at each of the options –

- A. $\int_0^s t dt = s^2/2$
- B. $\int_0^s 2t dt = 2s^2/2 = s^2$
- C. $\int_0^s t^2 dt = s^3/3$
- D. $\int_0^s (t + 1) dt = s^2/2 + s = s(s/2 + 1)$
- E. $\int_0^s 1 dt = s$.

So the correct answer is Option B.

[2]

Solution 30: Correct Answer is Option C.

We will derive this using first principles.

$${}_{t+h}p_x^{23} = {}_t p_x^{21} {}_h p_{x+t}^{13} + {}_t p_x^{22} {}_h p_{x+t}^{23} + {}_t p_x^{23} {}_h p_{x+t}^{33} + {}_t p_x^{24} {}_h p_{x+t}^{43}$$

$${}_h p_{x+t}^{13} = 0 \text{ and } {}_h p_{x+t}^{43} = 0$$

$${}_{t+h}p_x^{23} = {}_t p_x^{22} {}_h p_{x+t}^{23} + {}_t p_x^{23} {}_h p_{x+t}^{33} \dots\dots\dots (1)$$

$${}_h p_{x+t}^{33} = 1 - {}_h p_{x+t}^{32} - {}_h p_{x+t}^{34} \dots\dots\dots (2)$$

Using (1) and (2),

$${}_{t+h}p_x^{23} = {}_t p_x^{22} {}_h p_{x+t}^{23} + {}_t p_x^{23} (1 - {}_h p_{x+t}^{32} - {}_h p_{x+t}^{34}) \dots\dots\dots (3)$$

$${}_h p_{x+t}^{ij} = h * \mu_{x+t}^{ij} \dots\dots\dots (4)$$

From (3) and (4)

$${}_{t+h}p_x^{23} - {}_t p_x^{23} = {}_t p_x^{22} h \mu_{x+t}^{23} - {}_t p_x^{23} h \mu_{x+t}^{32} - {}_t p_x^{23} h \mu_{x+t}^{34}$$

$$({}_{t+h}p_x^{23} - {}_t p_x^{23}) / h = {}_t p_x^{22} \mu_{x+t}^{23} - {}_t p_x^{23} \mu_{x+t}^{32} - {}_t p_x^{23} \mu_{x+t}^{34}$$

$$d/dt ({}_t p_x^{23}) = \lim_{h \rightarrow 0+} ({}_{t+h}p_x^{23} - {}_t p_x^{23}) / h = {}_t p_x^{22} \mu_{x+t}^{23} - {}_t p_x^{23} \mu_{x+t}^{32} - {}_t p_x^{23} \mu_{x+t}^{34}$$

[3]

Solution 31: Correct Answer is Option A.

$$\text{Let } Y_t = X_t - X_{t-1}$$

$$\text{So, } Y_t = \alpha Y_{t-1} + e_t$$

$$\text{Cov}(Y_t, e_t) = \text{Cov}(aY_{t-1} + e_t, e_t) = \sigma^2.$$

$$\gamma_1 = \text{Cov}(Y_t, Y_{t-1}) = \text{Cov}(aY_{t-1} + e_t, Y_{t-1}) = a\gamma_0$$

$$\rho = \gamma_1 / \gamma_0 = a\gamma_0 / \gamma_0 = a$$

Estimate of α

= Sample autocorrelation function

$$= \sum_{i=2}^{20} (x_i - x_{t-1}) (x_{t+1} - x_t) / \sum_{i=2}^{20} (x_i - x_{t-1})^2$$

$$= 3 / 4$$

$$= 0.75$$

Using the Box-Jenkins methodology, forecast of x_{21} is given by:

$$x_{21} - x_{20} = 0.75(x_{20} - x_{19}) + 0$$

$$x_{21} = 39.2\% + 0.75(39.2\% - 42.1\%) = 37.025\%.$$

[2]

Solution 32: Correct Answer is Option C.

For deciding a better, we need to check that level of age setback at which the weighted sum of least squares i.e. M is minimum. In the present case it is minimum at $k = 3$. Hence, the ideal setback would be 3 years.

[2]

Solution 33: Correct Answer is Option D.

Let μ_1 be the force of weight loss in the first 30 days and μ_2 be the force of weight loss during the next 60 days.

$${}_{30}p_0 = 1 - 0.20 = 0.80$$

$$0.80 = \exp(-\mu_1 * 30)$$

$$\mu_1 = -\ln 0.80 / 30 = 0.007438 = 7.438 \text{ per 1,000}$$

$${}_{90}p_0 = 0.80 - 0.16 = 0.64$$

$${}_{90}p_0 = {}_{30}p_0 * {}_{60}p_{30}$$

$${}_{60}p_{30} = 0.64/0.80 = 0.80$$

$$0.80 = \exp(-\mu_2 * 60)$$

$$\mu_2 = -\ln 0.80 / 60 = 0.003719 = 3.7191 \text{ per 1,000}$$

[2]

Solution 34: Correct Answer is Option E.

$$Y_t$$

$$= \alpha * Y_{t-1} + Z_t$$

$$= \alpha * (\alpha * Y_{t-2} + Z_{t-1}) + Z_t$$

$$= \alpha^2 Y_{t-2} + \alpha Z_{t-1} + Z_t$$

$$= \alpha^2 (\alpha * Y_{t-3} + Z_{t-2}) + \alpha Z_{t-1} + Z_t$$

$$= \alpha^3 Y_{t-3} + \alpha^2 Z_{t-2} + \alpha Z_{t-1} + Z_t$$

$$= \sum_{i=0}^{\infty} \alpha^i Z_{t-i}$$

$$E(Y_t) = E(\sum_{i=0}^{\infty} \alpha^i Z_{t-i}) = 0$$

(as mean of each of the terms of the white noise process is 0 and hence their sum will also have a zero mean)

$$\begin{aligned} \text{Var}(Y_t) &= \text{Var}(\sum_{i=0}^{\infty} \alpha^i Z_{t-i}) \\ &= \sum_{i=0}^{\infty} \alpha^{2i} \text{Var}(Z_{t-i}) \\ &= \sum_{i=0}^{\infty} \alpha^{2i} \sigma^2 \\ &= \sigma^2 * (1 + \alpha^2 + \alpha^4 + \dots) \\ &= \sigma^2 / (1 - \alpha^2) \\ &= 48 / (1 - 0.5^2) \\ &= 48 / 0.75 \\ &= 64. \end{aligned}$$

Y_t is a sum of Z_t variables where $Z_t \sim N(0, \sigma^2)$.

Hence Y_t is also normally distributed.

$$\begin{aligned} Y_t &\sim N(E(Y_t), \text{Var}(Y_t)) \\ Y_t &\sim N(0, 64) \end{aligned}$$

$$\begin{aligned} P(Y_t > 10) &= P(Z > (10-0)/8) \\ &= P(Z > 1.25) = 1 - 0.89435 = 10.565\%. \end{aligned}$$

[3]

Solution 35: Correct Answer is Option B.

Let m_i denote the life in years of an air conditioner unit in state i (expected time till it reaches state 3 i.e. failure)

$$\lambda_1 = 0.02 + 0.05 = 0.07$$

$$m_1 = 1/0.07 + 0.02/0.07 * m_2 + 0.05/0.07 * m_3$$

We know $m_3 = 0$. As once dysfunctional, the unit does not have any life after reaching state 3.

$$m_1 = 100/7 + 2/7 * m_2 \dots\dots\dots (1)$$

$$\lambda_2 = 1 + 0.15 = 1.15$$

$$\begin{aligned} m_2 &= 1/1.15 + 1/1.15 * m_1 + 0.15/1.15 * m_3 \\ m_2 &= 100/115 + 100/115 * m_1 \dots\dots\dots (2) \end{aligned}$$

From (1) and (2),

$$\begin{aligned} m_1 &= 100/7 + 2/7 (100/115 + 100/115 m_1) \\ m_1 (1 - 2/7 * 100/115) &= 2/7 * 100/115 + 100/7 \\ m_1 &= 19.34 \text{ years i.e. 19 years.} \end{aligned}$$

[3]

Solution 36: Correct Answer is Option B.

$$E(X) = \Gamma(2 - 1/2) * \Gamma(1 + 1/2) * 2^{1/2} / \Gamma(2) = 0.5 * \sqrt{\pi} * 0.5 * \sqrt{\pi} * \sqrt{2} = \sqrt{0.125} \pi.$$

$$E(X^2) = \Gamma(2-1) * \Gamma(1+1) * 2 / \Gamma(2) = 1 * 1 * 2/1 = 2.$$

$$\text{Var}(X) = 2 - 0.125 \pi^2 \quad [2]$$

Solution 37: Correct Answer is Option **D**.

We need to use the Poisson thinning rule to solve this question.

Type 1 requests with response time of more than a minute = $50 / 10 = 5$

Type 2 requests with response time of more than a minute = $40 / 20 = 2$

Total requests with response time of more than a minute = 7. [2]

Solution 38: Correct Answer is Option **A**.

Under Gompertz Law, ${}_{10}p_{35}^G = g^{c^{35}(c^{10}-1)}$

Under Makeham's Law, ${}_{10}p_{35}^M = s^{10} * g^{c^{35}(c^{10}-1)}$

So, $0.75 = s^{10}$

After 10 years,

Under Gompertz Law, ${}_{10}p_{45}^G = g^{c^{45}(c^{10}-1)}$

Under Makeham's Law, ${}_{10}p_{45}^M = s^{10} * g^{c^{45}(c^{10}-1)} = 0.75 * g^{c^{45}(c^{10}-1)}$

$${}_{10}p_{45}^M = 75\% * {}_{10}p_{45}^G. \quad [2]$$

Solution 39: Correct Answer is Option **E**.

$$\log L(\mu) = d (\log \mu + \log E_x^C) - \mu E_x^C - \log d!$$

$$d/d\mu (\log L(\mu)) = d/\mu - E_x^C$$

$$d/d\mu^2 (\log L(\mu))^2 = -d / \mu^2$$

CRLB

$$= -1 / d/d\mu^2 (\log L(\mu))^2$$

$$= \mu^2 / d$$

$$= \mu * \mu / d$$

$$= \mu * d / (E_x^C * d)$$

$$= \mu / E_x^C \quad [2]$$

Solution 40: Correct Answer is Option **C**.

Gumbell copula has high upper tail dependence, no lower tail dependence.

Student's t copula has both upper and lower tail dependence.

Co-monotonic copula a high degree of positive interdependence throughout.

Clayton copula has high lower tail dependence, no upper tail dependence.

So, the right answer is A1-B4, A2-B3, A3-B1, A4-B2. [2]

Solution 41: Correct Answer is Option **A**.

Original Regression Model:

The regression coefficients β_1 and β_2 have been arrived in such a way that they minimize the TSS.

$$\begin{aligned} \text{TSS} &= \sum (Y - (\beta_1 X_1 + \beta_2 X_2))^2 \\ &= \sum (Y^2 - 2Y\beta_1 X_1 - 2Y\beta_2 X_2 + (\beta_1^2 X_1^2 + \beta_2^2 X_2^2 + 2\beta_1 X_1 \beta_2 X_2)) \end{aligned}$$

Taking the first order derivative with respect to β_1 and equating it with zero will give us the OLS estimate of β_1 .

$$\begin{aligned} d\text{TSS}/d\beta_1 &= -2\sum X_1 Y + 2\beta_1 \sum X_1^2 + 2\beta_2 \sum X_1 X_2 \end{aligned}$$

$$\begin{aligned} \text{Equating with 0,} \\ (\sum X_1^2)\beta_1 &= \sum X_1 Y - \beta_2 \sum X_1 X_2 \end{aligned}$$

Substituting values given in the question,

$$\begin{aligned} (10)(4) &= \sum X_1 Y - (3)(20) \\ \sum X_1 Y &= 40 + 60 = 100 \end{aligned}$$

It can be checked that the second order derivative with respect to β_1 is > 0 and hence its minima.

Taking the first order derivative with respect to β_2 and equating it with zero will give us the OLS estimate of β_2 .

$$\begin{aligned} d\text{TSS}/d\beta_2 &= -2\sum X_2 Y + 2\beta_2 \sum X_2^2 + 2\beta_1 \sum X_1 X_2 \end{aligned}$$

$$\begin{aligned} \text{Equating with 0,} \\ (\sum X_2^2)\beta_2 &= \sum X_2 Y - \beta_1 \sum X_1 X_2 \end{aligned}$$

Substituting values given in the question,

$$\begin{aligned} (12)(3) &= \sum X_2 Y - (4)(20) \\ \sum X_2 Y &= 80 + 36 = 116 \end{aligned}$$

It can be checked that the second order derivative with respect to β_2 is > 0 and hence its minima.

Ridge Regression Model:

The regression coefficients β_{R1} and β_{R2} have been arrived in such a way that they minimize the revised TSS.

$$\begin{aligned} \text{TSS}_R &= \sum (Y - (\beta_{R1} X_1 + \beta_{R2} X_2))^2 + \lambda * (\beta_{R1}^2 + \beta_{R2}^2) \\ &= \sum (Y^2 - 2Y\beta_{R1} X_1 - 2Y\beta_{R2} X_2 + (\beta_{R1}^2 X_1^2 + \beta_{R2}^2 X_2^2 + 2\beta_{R1} X_1 \beta_{R2} X_2)) + \lambda * (\beta_{R1}^2 + \beta_{R2}^2) \end{aligned}$$

Taking the first order derivative with respect to β_{R1} and equating it with zero will give us the estimate of β_{R1} .

$$\begin{aligned} dTSS_R/d\beta_{R1} \\ = -2\sum X_1Y + 2\beta_{R1}\sum X_1^2 + 2\beta_{R2}\sum X_1X_2 + 2\lambda\beta_{R1} \end{aligned}$$

$$\begin{aligned} \text{Equating with 0,} \\ (\sum X_1^2 + \lambda)\beta_{R1} = \sum X_1Y - \beta_{R2}\sum X_1X_2 \end{aligned}$$

$$\begin{aligned} \text{Substituting values given in the question,} \\ (10 + 5)\beta_{R1} = 100 - 20\beta_{R2} \\ 15\beta_{R1} = 100 - 20\beta_{R2} \dots\dots\dots (1) \end{aligned}$$

It can be checked that the second order derivative with respect to β_{R1} is > 0 and hence its minima.

Taking the first order derivative with respect to β_{R2} and equating it with zero will give us the estimate of β_{R2} .

$$\begin{aligned} dTSS_R/d\beta_{R2} \\ = -2\sum X_2Y + 2\beta_{R2}\sum X_2^2 + 2\beta_{R1}\sum X_1X_2 + 2\lambda\beta_{R2} \end{aligned}$$

$$\begin{aligned} \text{Equating with 0,} \\ (\sum X_2^2 + \lambda)\beta_{R2} = \sum X_2Y - \beta_{R1}\sum X_1X_2 \end{aligned}$$

$$\begin{aligned} \text{Substituting values given in the question,} \\ (15+5)\beta_{R2} = 116 - (20)\beta_{R1} \\ (20)\beta_{R2} = 116 - (20)\beta_{R1} \dots\dots\dots (2) \end{aligned}$$

It can be checked that the second order derivative with respect to β_{R2} is > 0 and hence its minima.

Solving equations (1) and (2) simultaneously,

$$\begin{aligned} \beta_{R1} &= 3.20 \\ \beta_{R2} &= 2.60. \end{aligned}$$

[3]
